

Figure 1 is a line plot showing the relative error of the static correlation energy, $E^{\text{static_err}}$, as a function of the number of occupied orbitals, N_{occ} . The x-axis represents N_{occ} and ranges from 2.5 to 5.5. The y-axis represents $E^{\text{static_err}}$ and ranges from 0.00 to 0.04. A vertical dashed line at $N_{\text{occ}} = 3.0$ indicates the equilibrium value. The legend identifies six data series: Equilibrium (dashed line), SAPT(PBE0)-D4 (S, ND) (purple triangles), SAPT(PBE0)-D4 (I, ND) (orange inverted triangles), SAPT(PBE0)-D4 (S) (red squares), SAPT(PBE0)-D4 (I) (green diamonds), and SAPT2+3(CCD)/aTZ (black circles). The ND series show a peak at $N_{\text{occ}} = 3.0$, while the other series show a minimum at $N_{\text{occ}} = 3.0$.

N_{occ}	Equilibrium	SAPT(PBE0)-D4 (S, ND)	SAPT(PBE0)-D4 (I, ND)	SAPT(PBE0)-D4 (S)	SAPT(PBE0)-D4 (I)	SAPT2+3(CCD)/aTZ
2.5	0.000	0.025	0.005	0.005	0.005	0.005
2.7	0.000	0.028	0.008	0.008	0.008	0.008
2.9	0.000	0.030	0.010	0.010	0.010	0.010
3.0	0.000	0.030	0.005	0.005	0.005	0.005
3.1	0.000	0.028	0.008	0.008	0.008	0.008
3.5	0.000	0.025	0.015	0.015	0.015	0.015
4.2	0.000	0.020	0.020	0.020	0.020	0.020
5.6	0.000	0.015	0.025	0.025	0.025	0.025

Figure 1 is a line graph showing the dispersion energy (in kcal/mol) as a function of the distance between the two ethene fragments (in Å) for the [2+2] cycloaddition of ethene and ethyne. The x-axis ranges from 2.5 to 5.5 Å, and the y-axis ranges from 0 to -9 kcal/mol. A vertical dashed line is drawn at 2.9 Å. The legend identifies six computational methods: SAPT(PBE0)-D4 (S) (red squares), SAPT(PBE0)-D4 (I) (green diamonds), HF-D4 (blue circles), SAPT(PBE0)/aDZ (brown crosses), SAPT(PBE0)/aTZ (grey plus signs), and SAPT2+3(CCD)/aTZ (black circles). All methods show a sharp increase in dispersion energy as the distance decreases below 3.0 Å, with the SAPT2+3(CCD)/aTZ method showing the highest dispersion energy at the smallest distances.

Distance (Å)	SAPT(PBE0)-D4 (S)	SAPT(PBE0)-D4 (I)	HF-D4	SAPT(PBE0)/aDZ	SAPT(PBE0)/aTZ	SAPT2+3(CCD)/aTZ
2.5	-9.0	-8.8	-7.2	-7.0	-7.8	-8.2
2.7	-7.0	-6.8	-5.5	-5.5	-5.8	-6.2
2.9	-4.0	-3.8	-3.5	-3.5	-3.8	-4.2
3.1	-3.2	-3.0	-2.8	-2.8	-3.0	-3.5
3.5	-1.8	-1.6	-1.5	-1.5	-1.6	-1.8
4.2	-0.8	-0.6	-0.5	-0.5	-0.6	-0.8
5.6	-0.2	-0.1	-0.1	-0.1	-0.2	-0.3

Distance (Å)